

Game Design Document

Liminophobia: The Forgotten Space

"Survive the unknown. Find your way out"

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2. Game Overview

2.1 Game Concept

This is a *Backrooms* inspired 3D survival horror game. The player will wake up in a mysterious environment and must explore a series of levels while completing objectives and interacting with NPCs. Along the way the player must manage their resources, solve puzzles, avoid entities, and find a way to escape. What makes the game interesting is as the player progresses it gets harder and there are less hiding spots which force the player to think of different strategies to progress the levels.

2.2 Target Audience

This game is intended for players aged 15 and older who enjoy horror, exploration, and puzzle solving. The game is designed for players who are drawn to suspense and mystery rather than constant combat. (e.g., someone who enjoys the tension of not knowing what is around the corner more than the payoff of fighting it)

2.3 Genre

This game is primarily a survival horror and exploration game. It also incorporates puzzle solving and stealth mechanics. Tension is generated through exploration of unsettling liminal spaces, limited resources, and unpredictable entity encounters rather than direct combat.

3. Gameplay & Mechanics

3.1 Gameplay

3.1.1 Game Progression

The player advances through a tutorial level followed by four main levels. Progression is level-based and one directional. Each level must be completed before the next unlocks. As the player advances, entities become more aggressive and available hiding spots decrease, which is the primary way the game escalates difficulty from one level to the next.

3.1.2 Mission/Challenge/Puzzle Structure

Each level (other than the tutorial) centers on a main quest given by a friendly NPC that the player must complete to progress. Alongside this, NPCs offer trades and an optional NPC-rescue side quest is available after the main objective is complete, rewarding the player with a special item on success.

3.1.3 Objectives

The player's main objective is to survive long enough to escape the liminal dimension. On a per-level basis, this means completing the NPC's quest, avoiding or evading hostile entities, and reaching the level's exit.

3.1.4 Play Flow

A typical play session flows as: Explore the current area → Interact with NPCs or objects → Solve any puzzles and complete quest steps → Avoid or hide from entities as needed → Reach the level exit. If the player dies, they respawn at the start of the level they died on rather than losing overall progress, keeping the flow moving forward instead of punishing the player with a full restart.

3.2 Mechanics

3.2.1 Rules

Explicit rules include:

- The player has a limited stamina pool that depletes while sprinting and regenerates while walking or standing still
- The inventory has limited capacity
- Hiding spot availability decreases by roughly 15% with each level
- There is no permadeath (Dying respawns the player at the start of the current level)

Implicit rules include:

- Entities react to the player's noises (running, interacting)
- Escalating from patrol to pursuit when they detect the player

3.2.2 Physics

The game uses standard human scale 3D physics. Thus, gravity, solid collision with walls and objects, and no special player abilities such as double jumping or wall climbing. Movement is deliberately grounded and realistic to reinforce the liminal realism tone.

3.2.3 Character Movement

The player moves using WASD, looks around using the mouse, and can hold Shift to sprint at the cost of stamina. There is no crouch or jump mechanic in the current design. Movement options are intentionally minimal to keep the focus on tension and decision making rather than movement tech

3.2.4 Actions

The player presses E to interact with objects, NPCs, and pickable items, and Left Click to toggle the flashlight. The Esc key opens the pause menu. Picked up items go into the inventory and can be selected using number key hotkeys shown in the HUD.

4. Characters

4.1 Characters

The Player

Short Backstory: An ordinary person with no memory of how they ended up in the Backrooms. No prior history is revealed to the player.

Personality: Not predefined. The player's personality is expressed through the choices made in dialogue, whether they succeed in NPC rescues, and how they approach entities.

Appearance: An unremarkable, ordinary looking guy in plain clothes.

Abilities: Can walk, sprint (limited by stamina), interact with objects and NPCs, use a flashlight, and carry a limited inventory. Has no combat abilities.

Relationship to Other Characters: Start as a stranger to every NPC and entity in the world. Relationships with NPCs can improve through successful rescues.

4.1.1 Friendly Characters

Sentient Bottle of Yager (Tutorial NPC)

Short Backstory: A bottle of Yager somehow gained consciousness after being drunk and left behind in the Backrooms; Found sitting in the Puzzle Room, it is the first "person" the player meets

Personality: Chatty, upbeat, and a little unhinged (treats the horror around it with dark humor, which keeps the tutorial's tone light before the game turns tense)

Appearance: A dusty, empty bottle of Yager

Abilities: Can talk and hand out a quest. Has no physical form beyond the bottle, so it cannot move, follow the player, or fight.

Relationship to Other Characters: Friendly to the player from the very first interaction. Does not interact with other characters since it is stationary.

Nebula (Level 1 NPC)

Short Backstory: A small, cat shaped void creature that somehow ended up trapped in the Backrooms alongside everyone else.

Personality: Skittish around strangers at first, but affectionate once the player earns its trust.

Appearance: A cat shaped silhouette with faint, shifting starlight patterns where fur would be.

Abilities: Must pet using a special pair of gloves. (protecting the player's hands from its void like texture) (**PLAYER DIES IF THEY PET WITHOUT GLOVES**) Once trust is built, Nebula hands the player a quest

Relationship to Other Characters: Bonds specifically with the player through a petting mechanic. Does not interact with other NPCs or entities

The Speedster (Level 2 NPC)

Short Backstory: A former speedster who became a chronic alcoholic. Now just lives in the Backroom as a sad drunk.

Personality: Self-aware and slurs words. Sometimes Paranoid.

Appearance: A disheveled man with wrinkled/stained clothes, bloodshot eyes and a half empty bottle he clutches

Abilities: Runs a trade system (one time offer), exchanging a booze type item for a decent item.

Relationship to Other Characters: Bonds with the player if they are given a booze item. Neutral to all NPCs.

The Traveler (Level 3 NPC)

Short Backstory: A wanderer who has been moving through the Backrooms far longer than anyone else, trading whatever he finds for whatever he needs. No one knows how long he has actually been here for.

Personality: Calm and transactional. Gives nothing for free, and speaks like he has already seen everything the Backrooms has to offer

Appearance: Wrapped head to toe in a black robe, hood pulled forward far enough that his face is never visible.

Abilities: Runs a trade system (one time offer), exchanging two items for one good/decent item.

Relationship to Other Characters: Neutral toward the player and every other NPC

Server (Level 4 NPC)

Short Backstory: Not a character in the traditional sense. (An old mechanical terminal wired into a heavy, mechanically locked door)

Personality: N/A. Non sentient, though it may play simple automated status/error sounds when modules are inserted correctly or incorrectly

Appearance: A wall mounted terminal with four module slots, humming and flickering with age.

Abilities: Accepts four modules scattered around the level; Once all four are inserted, it unlocks the mechanical door blocking the final exit

Relationship to Other Characters: None (Purely a gating mechanic rather than a relationship based character)

4.1.2 Hostile Characters

Entity 0565 (Level 4 Hostile)

Short Backstory: Once something living (the deer skull and antlers are the only recognizable remains)

Personality: Purely instinctual. Stalks quietly until it has a clear target then commits fully to the chase rather than giving up or wandering off

Appearance: A hunched, quadrupedal creature that moves low to the ground on dark, sinewy limbs with a bleached deer like skull and antlers for a head.

Abilities: Low profile quadrupedal movement makes it hard to spot at range. Once it detects the player it commits to a fast, sustained pursuit rather than losing interest quickly, so breaking line of sight matters more than outrunning it

Relationship to Other Characters: Purely antagonistic toward the player. Does not interact with friendly NPCs unless rescue where it is free to attack NPC

The Damned (Level 2 Hostile)

Short Backstory: A fleshy, mangled monster of unclear origin

Personality: Driven by instinct rather than intent. Does not hunt so much as react violently to anything nearby

Appearance: A mangled, fleshy mass, unsettling more for its wrongness than any specific monstrous feature.

Abilities: Slower than the player but persistent and hard to lose once it locks on, forcing the use of hiding spots rather than outrunning it

Relationship to Other Characters: Purely antagonistic toward the player. Does not interact with friendly NPCs unless rescue where it is free to attack NPC

The Shadow (Level 3 Hostile)

Short Backstory: A hostile presence with no clear physical form, native to the Backrooms.

Personality: Eerie and unpredictable. Its lack of a physical form makes its behavior feel less rule-bound than other entities

Appearance: A shifting, shadow-like figure with no solid silhouette (harder to pin down visually than the other entities)

Abilities: Detection and pursuit behave like other entities, but its indistinct form makes it harder for the player to tell exactly where it is at a glance, especially in dim lighting

Relationship to Other Characters: Purely antagonistic toward the player. Does not interact with friendly NPCs unless rescue where it is free to attack NPC

Bonewalker (Level 1 Hostile)

Short Backstory: A stitched together mess of animal and human bones, assembled by unknown means into something that can move

Personality: Puppet-like and unnatural (does not behave like a living creature so much as something being dragged upright and forced to walk)

Appearance: A stitched together collection of animal and human bones that clatters audibly as it drags itself along, moving like a pupper with half its strings cut

Abilities: Its clattering movement gives the player an audio cue to detect it before it detects them, making sound awareness especially important against this entity

Relationship to Other Characters: Purely antagonistic toward the player. Does not interact with friendly NPCs unless rescue where it is free to attack NPC

5. Levels

Liminophobia currently consists of one tutorial level followed by four main levels, described below:

5.1 Level Overview

Level	Description
Tutorial	Explains the basics of this game, including hiding and how items work. THE PLAYER CAN NOT DIE ON THIS LEVEL
Level 1	The first real level. The player can die here, which restarts the level. The player has to complete the quest by the NPC to move on. Minus 15% chance of hiding spot
Level 2	Gets harder than Level 1: Entities are more aggressive and hiding spots continue to thin out and aggro range gets increased
Level 3	Continues the escalation in enemy aggression and puzzle complexity
Level 4	The last and hardest level, with the fewest hiding spots and most aggressive entities. Entities can now hear the flashlight being turned on and off

5.2 Training Level

5.2.1 Synopsis

The player wakes up alone in a small, enclosed section of the Backrooms with no immediate threats present. A short guided sequence introduces the core systems before the player is allowed into the rest of the game

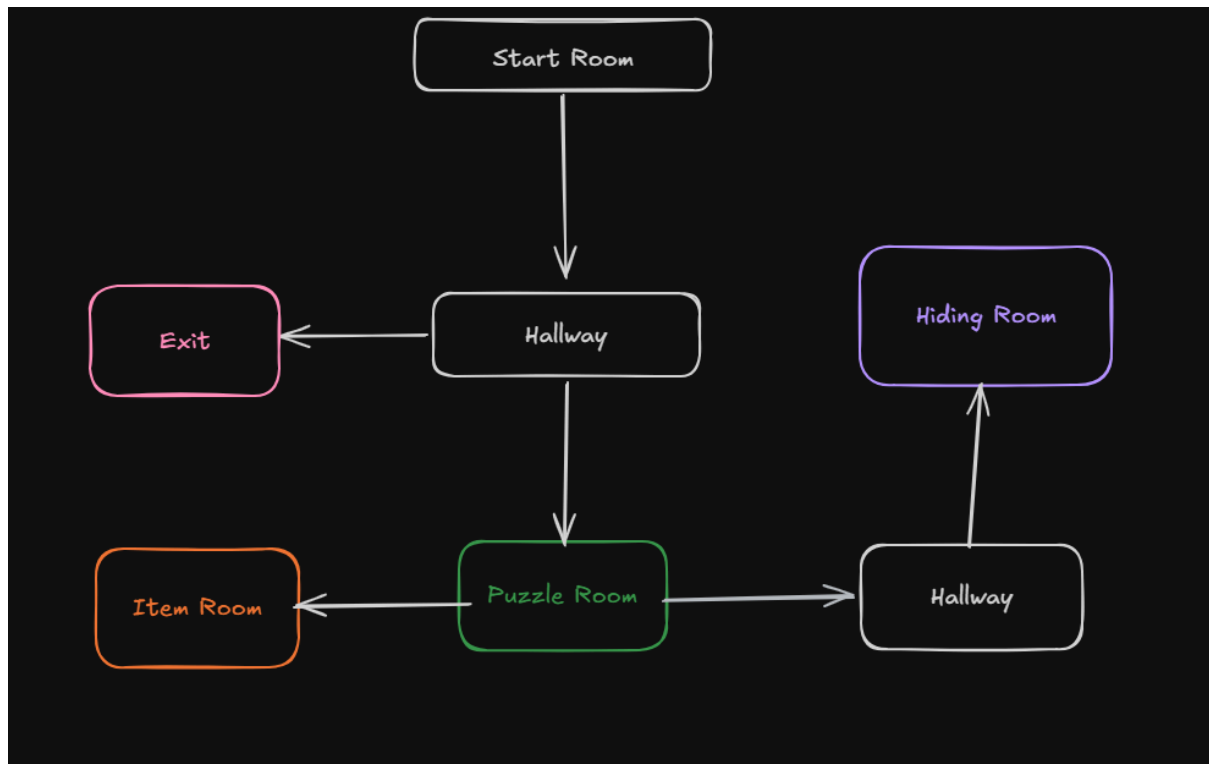
5.2.2 Objectives

Learn to move and interact with the world, pick up and use an item, use a hiding spot and locate the level exit

5.2.3 Details of What Happens in The Level

The player starts in the Start Room and moves into the Hallway, where they find an NPC if they go straight else they find a locked Exit door. (Note the player won't know that's the exit room until Quest given to find exit and they interact with that door) Talking to the NPC triggers a short dialogue with a few selections based on what the player chooses; One of which is how to get out, and the NPC admits they do not know for certain, only that there's a room somewhere that might help. This exchange creates the level's first quest. When the player goes back and tries the Exit door, it does not open, and the objective updates to "Find the key". This sends the player further into the level through the Puzzle Room, where the player can then choose to go to the Item Room or through a Hallway which then allows the player to go to the Hiding Room which is used to teach the hiding mechanic. Once the player has the key, they backtrack to the first Hallway and unlock the Exit door. Then once completed the player can enter the Exit Room but there's a missing item that is needed in order to fix the door to leave. Thus, this teaches the player to explore. Once the player finds the items needed they can fix the door and the level is complete.

5.2.4.1 Map



5.2.4.2 Critical Path

Start Room → Hallway → Puzzle Room → Item Room → Puzzle Room → Hallway → Exit Room → Hallway → Puzzle Room → Second Hallway → Hiding Room → Second Hallway → Puzzle Room → Hallway → Exit Room → Exit Door

5.2.4.3 Important & Incidental Encounters

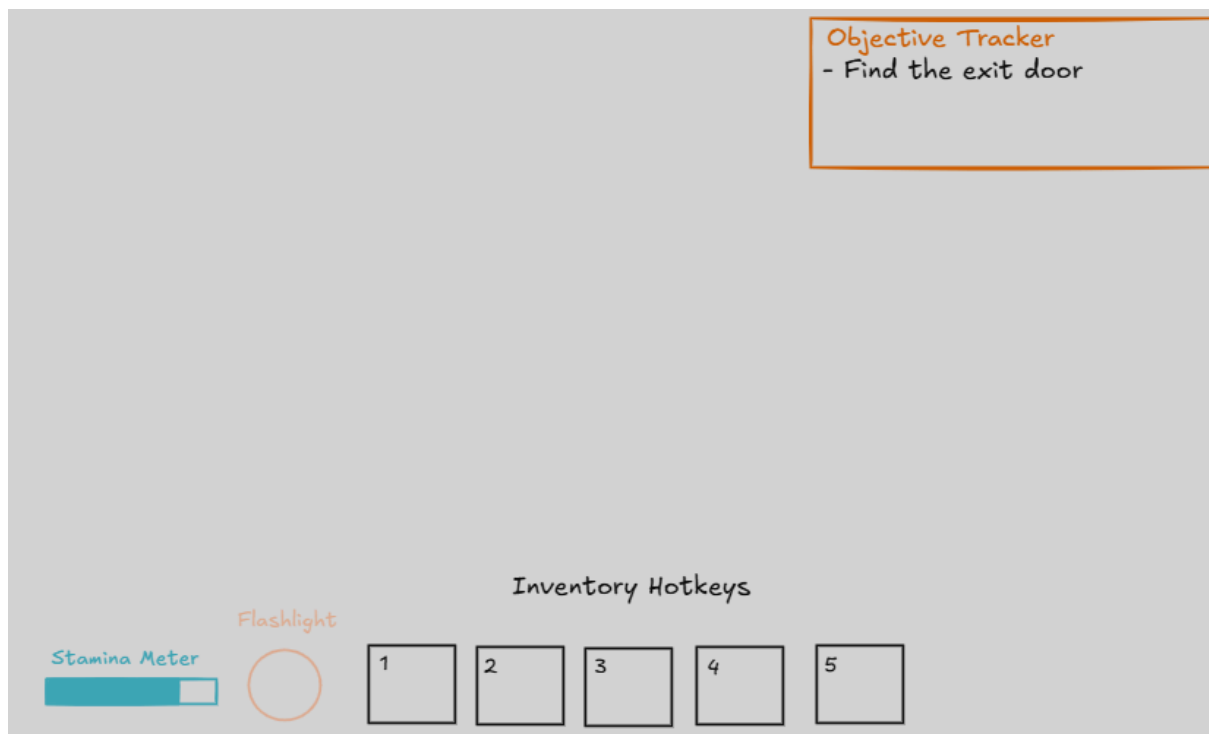
The key encounter is the NPC dialogue in the Puzzle Room: the NPC does not hand the player a direct answer, only a vague lead, which is what generates the quest and pushes the player to explore rather than backtrack immediately. The locked Exit door is a soft-fail moment that redirects the objective from “Find a way out” to “Find the key” once the player tries it (Only true if the player has the quest “Find a way out”).

6. Interface

6.1 Visual System

6.1.1 HUD

The on-screen display includes a stamina meter (Bottom left corner), a quest/objective tracker (Top right corner), and inventory hotkeys (Bottom center) for a quick item selection. A flashlight battery status indicator sits to the right of the Stamina bar. **SEE THE MOCKUP BELOW**



6.1.2 Menus

Pressing Esc opens the pause menu, which is expected to include Resume, Options, and Quit. Inventory is handled as its own full screen menu using a size based grid where each item can take up one, two, three, or four cells depending on its size. From this grid, the player can assign up to 5 items to the HUD hotkeys for quick access during gameplay

6.2 Control System

<i>Input</i>	<i>Action</i>
WASD	Movement
Mouse	Camera
E	Interact
Shift	Run
Esc	Pause Menu
Left Click	Flashlight
I	Inventory

6.3 Audio, Music, Sound Effects

Background Music: Quiet, ambient music plays while exploring, becoming more intense when the player encounters or is chased by an entity by layering in screams and distorted noises over the ambient track. Also, background noises when hiding

References:

[Horror Background Atmosphere 026 \(30\) by Universfield | Royalty-free Music - Pixabay](#)

[Download Free Bass Sound Effects | Mixkit](#)

[Scream Horror SFX by Jusatti890 | Royalty-free Music - Pixabay](#) (Level 2 Entity)

[Scream Horror SFX by Jusatti890 | Royalty-free Music - Pixabay](#) (Level 2 Entity)

[Corrupt scream by CANNIBAL REN | Royalty-free Music - Pixabay](#) (Level 4 Entity)

[Heart Beat by SoundReality | Royalty-free Music - Pixabay](#) (Hiding background)

Sound Effects: Player footsteps, entity sounds, doors opening and closing, flashlight sounds, picking up and dropping items, interacting with objects, NPC sounds and dialogue, running sounds , and environmental sounds such as buzzing lights and distant noises

References:

[chain by freesound_community | Royalty-free Music - Pixabay](#) (Entity 1)
[Creepy laugh Sound by u_c98kb8pdp9 | Royalty-free Music - Pixabay](#) (Entity 3)
[Keyboard Typing SFX by SoundReality | Royalty-free Music - Pixabay](#) (NPC talk)
[Fah! by JohnnyBacon156 | Royalty-free Music - Pixabay](#) (Death Sound)
[walking up the stairs by FreeEverythingxx | Royalty-free Sound Effects - Pixabay](#)
(Player Footsteps)
[garage door opening by flutie8211 | Royalty-free Sound Effects - Pixabay](#) (NPC 4)
[Opening Door by SoundReality | Royalty-free Sound Effects - Pixabay](#) (Creak free)
[Open Door Stock SFX by DRAGON-STUDIO | Royalty-free Sound Effects - Pixabay](#)
(Creak)
[Door Slam by Universfield | Royalty-free Sound Effects - Pixabay](#) (Slam)
[Closed Door by SoundReality | Royalty-free Sound Effects - Pixabay](#) (Slam Free)
[Bubble Pops by Homemade_SFX | Royalty-free Sound Effects - Pixabay](#) (Drop)
[Fluorescent light buzzing by freesound_community | Royalty-free Sound Effects - Pixabay](#) (Buzzing Light)
[Disturbing Glitch Monsters 1 by XtremeFreddy | Royalty-free Sound Effects - Pixabay](#) (Post Death Scene)
[Static Noise by u_xg7ssi08yr | Royalty-free Sound Effects - Pixabay](#) (Death Noise)

7. Artificial Intelligence

Liminophobia uses simple finite-state AI (patrol → detect → pursue → lose interest) rather than complex learning based behavior, which keeps enemy behavior predictable enough to design around while still feeling threatening.

7.1 Opponent & Enemy AI

Hostile entities patrol assigned areas of each level and switch into a hunting state when they detect the player through sound. Once alerted, an entity pursues the player until it loses them. Typically because the player breaks line of sight or hides in time. Aggression, detection range, and pursuit persistence all scale up with each level, which is what drives the game's escalating difficulty rather than simply adding more enemies.

7.2 Non-Combat & Friendly Characters

Friendly NPCs remain in fixed locations or patrol small safe areas and initiate dialogue when the player interacts with them. Their available dialogue, quests, and trade options change based on the player's prior choices. For example, whether an earlier NPC rescue succeeded or failed giving the sense that NPCs remember and react to the player's playthrough.

8. Game Art

8.1 Look & Feel

The game uses Backroom entities that exist within the multiple versions of Backrooms lore. It is strictly 3D, since the game wouldn't be the same if it was in 2D. The style described as liminal realism is the feeling of being between the human and realms beyond human comprehension, blending hyperreality and abstraction.

1. **Visual Reference:**

a. Backrooms: Escape Together



b. Escape The Backrooms



c. Observation Duty

